MicrOSAL — Backend Capability Matrix

FreeRTOS	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	cfg
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✓
stack_overflow_detection	cfg
cpu_load_stats	cfg
wait_set	✗
native_event_flags	✓
native_work_queue	✗
native_condvar	✗
native_memory_pool	✗
native_stream_buffer	✓
isr_stream_buffer	✓
native_message_buffer	✓
isr_message_buffer	✓
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

Zephyr	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✓
timed_join	✓
thread_suspend_resume	✓
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✓
cpu_load_stats	✓
wait_set	✗
native_event_flags	✓
native_work_queue	✓
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✓
barrier	✓
monotonic_clock	✓

Linux	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✗
thread_affinity	✓
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✓
native_event_flags	✗
native_work_queue	✗
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✓
spinlock	✗
barrier	✓
monotonic_clock	✓

POSIX	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✗
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✓
native_event_flags	✗
native_work_queue	✗
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✓
spinlock	✗
barrier	✓
monotonic_clock	✓

RTEMS	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✗
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✓
native_event_flags	✗
native_work_queue	✗
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✓
spinlock	✗
barrier	✓
monotonic_clock	✓

INTEGRITY	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✗
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✓
native_event_flags	✗
native_work_queue	✗
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✓
spinlock	✗
barrier	✓
monotonic_clock	✓

ThreadX	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✓
native_work_queue	✗
native_condvar	✗
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

VxWorks	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✗
native_work_queue	✓
native_condvar	✓
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

NuttX	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✗
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✓
cpu_load_stats	✗
wait_set	✗
native_event_flags	✗
native_work_queue	✓
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✓
spinlock	✗
barrier	✓
monotonic_clock	✓

QNX	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✗
thread_affinity	✓
timed_join	✓
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✗
native_work_queue	✗
native_condvar	✓
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✓
spinlock	✗
barrier	✓
monotonic_clock	✓

ChibiOS	
recursive_mutex	✗
timed_mutex	✗
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✓
native_work_queue	✗
native_condvar	✓
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

Bare-metal	
recursive_mutex	✗
timed_mutex	✗
priority_inheritance	✗
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✗
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✗
native_work_queue	✗
native_condvar	✗
native_memory_pool	✗
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

PX5	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✓
native_event_flags	✓
native_work_queue	✗
native_condvar	✗
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

Micrium	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✓
native_work_queue	✗
native_condvar	✗
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

embOS	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✓
stack_overflow_detection	✓
cpu_load_stats	✓
wait_set	✗
native_event_flags	✓
native_work_queue	✗
native_condvar	✗
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

CMSIS-RTOS	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✗
native_work_queue	✗
native_condvar	✗
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓

CMSIS-RTOS2	
recursive_mutex	✓
timed_mutex	✓
priority_inheritance	✓
isr_semaphore	✓
thread_affinity	✗
timed_join	✗
thread_suspend_resume	✗
thread_local_data	emu
dynamic_thread_priority	✓
task_notification	✗
stack_overflow_detection	✗
cpu_load_stats	✗
wait_set	✗
native_event_flags	✓
native_work_queue	✗
native_condvar	✗
native_memory_pool	✓
native_stream_buffer	✗
isr_stream_buffer	✗
native_message_buffer	✗
isr_message_buffer	✗
native_rwlock	✗
spinlock	✗
barrier	✓
monotonic_clock	✓